

José Manuel Requena Plens

Firmware & Software Engineer Firmware • Software • QA

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Embedded firmware and software engineer with experience building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid R&D background. Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

// EXPERIENCE

R&D Software Developer

Apr 2023 - May 2026

Power Electronics S.L. • Solar + Storage • Llíria, Spain

Firmware development for solar inverters and control systems in photovoltaic plants, working on STM32 microcontrollers with bare-metal and RTOS environments.

- Reduced RTOS task stack usage by 30%–50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50%.
- Implemented a scheduled purge (daily/monthly) that safely clears thousands of files, preventing storage exhaustion under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: ~50–80 unit tests per module (Ceedling, thousands in total) reaching 100% line and branch coverage via standard-library mocks/fakes, plus hundreds of on-hardware integration tests (pytest) — TCP/FTP performance and FTP security —; all in CI.
- Reviewed merge requests across a 12-person team, catching bugs and proposing improvements before integration.

Predoctoral researcher

Jan 2023 - Apr 2023

Polytechnic University of Valencia + Spanish National Research Council + Hospital La Fe • Medical Imaging and Therapy Systems • Valencia, Spain

Researcher in the R&D line: *Acoustic Metamaterials for Ultrasound Histotripsy treatments*.

- Ultra-close focus acoustic lens design.
- Prototype manufacturing.
- Experimentation with ex-vivo tissues.

Predoctoral researcher

Jul 2021 - Jul 2022

Polytechnic University of Valencia + Spanish National Research Council • Medical Imaging and Therapy Systems • Valencia, Spain

The main objective is to conduct research and development of acoustic metamaterials for use in architectural applications and medical imaging.

- Developed acoustic metadiffusers reaching a diffusion coefficient of 0.8 with a thickness 10× smaller than traditional diffusers.
- Simulation of the acoustic diffusers and validation of results (MATLAB & COMSOL).
- Designed and 3D-printed (resin) slitted-tube waveguides with numerically-computed slits (2–3 cm tubes, ~0.5 mm slits) that increased ultrasound directivity and range ~15× in air, validated experimentally.
- Validated experimental results against those obtained in simulation.
- Advisor of the Master thesis of a student of the Master's Degree in Acoustic Engineering of the UPV.

Researcher in European project

Feb 2020 - Jul 2021

Higher Polytechnic School of Gandia + European Space Agency • Complex Media Acoustics Group of IGIC • Gandia, Spain

TECHNICAL SKILLS

Embedded Firmware & Electronics

C
STM32 & ARM
ESP32 / ESP-IDF
PlatformIO
Texas Instruments
FreeRTOS / RTOS
Comm. Protocols
HW Debugging
Keil uVision
Instrumentation
Secure Boot & Embedded Security
Doxygen



Software Engineering & Tools

Python
C++
Go
TypeScript
Bash
Clang Tooling
Git
GitHub
GitLab
Docker
MATLAB
CMake
JetBrains Toolbox



Quality, CI/CD & DevSecOps

GitHub Actions
GitLab CI
Jenkins
SonarQube / SonarCloud
GoReleaser
Static Analysis & SAST
Testing
pre-commit hooks



Networking

TCP/IP & Networking
MikroTik RouterOS
WireGuard / VPN
IPv6 & Dual-stack
QUIC / HTTP3
DNS / Dynamic DNS



Security & Cryptography

Applied Cryptography
TLS / mTLS / PKI
Nginx Hardening
CrowdSec
Network Hardening



AI & Agentic Engineering

Model Context Protocol (MCP)
AI Coding Agents
LLM Integration



Simulation & Acoustics

COMSOL Multiphysics
LabView



Implement and experimentally validate a sound mitigation method, applicable to a real space vehicle launch configuration (VEGA), that results in a significant decrease in sound pressure levels generated in the launch area during spacecraft liftoff.

- Designed the optimum acoustic-metamaterial geometry, achieving near-total absorption ($\alpha \approx 1$) across 200–500 Hz with a subwavelength design (wavelength/thickness ratio ≈ 16.5 vs ~ 2 conventional), validated on real lab prototypes.
- Simulation of a real environment to validate design performance (MATLAB, COMSOL & Python).
- Designed and 3D-printed the first prototype; for the large-scale version designed the injection-molding model and sourced the manufacturing of 600 modules with a supplier.
- Development of software for acoustic measurements according to ISO 10534-2 and ASTM E2611 standards. [A|Lab](#)
- Design and development of a 3-axis robotic system to perform experimental measurements. Photos and some information here: [NEWS](#)
- Results validated and accepted by ESA.

Research internship

Feb 2018 - Jul 2018

[University of Alicante](#) · [Dep. of Physics, Systems Engineering and Signal Theory](#) · Alicante, Spain

Internship in different research projects in the [Applied Acoustics Group](#), belonging to the University Institute of Physics Applied to Sciences and Technologies of the Higher Polytechnic School of Alicante.

- Simulation of acoustic models.
- Acoustic measurements by means of a LabView-controlled robotic system.
- Investigation of the acoustic radiation efficiency (Vibration-borne noise) of metallic plates for an initial study of ship noise emissions in water. Carried out in collaboration with [SAES](#), here you can read a news item on the subject: [NEWS](#).

Founder and Vocal Member

Sep 2016 - Jul 2018

[AmicroTech](#) · [University of Alicante](#) · Alicante, Spain

Association based at the Higher Polytechnic School of the University of Alicante. Promotes project-oriented microcontroller programming knowledge.

- Taught 30+ students in reinforcement and extension classes for the subject "Digital Electronic Systems" of the Degree in Sound and Image Engineering of the University of Alicante.
- Laboratory organization and planning (located at the Colegio Mayor).
- Communication and Public Relations.

Sound technician

Jun 2011 - Feb 2018

[Acusticox](#) · Cox, Spain

- Equipment installation and adjustment for events.
- Equipment installation and adjustment for permanent installation.
- FOH Technician.
- Customer acquisition.

Laboratory technician

Sep 2005 - Nov 2009

[Profdent](#) · Alicante, Spain

- Reception of work and casting of plaster models from impressions (alginate/silicone), with and without implants.
- Mounting articulators and casting metal crowns in a centrifuge.
- Ceramic application on crowns and bridges.

// OPEN SOURCE PROJECTS

gitlab-mcp-server

Go · Model Context Protocol server

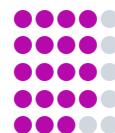
30 ★ · 50 releases · ~130k downloads · Listed on Glama, mcp.so, Cursor & mcpServers.org
Open-source GitLab MCP server that exposes 850+ GitLab actions (1,000+ on Enterprise) to AI assistants through a dynamic 2-tool find/execute design, with stdio/HTTP/OAuth transports and safe & read-only modes.

Acoustic Instrumentation



General & Platforms

Linux / Unix
MacOS
Windows
LaTeX
HTML / CSS



LANGUAGES

Spanish
English

Native
Professional

📍 On-site, remote or hybrid

CERTIFICATES

Occupational Health and Safety

[Generic](#) (INVASSAT, 50 h)
[Nanomaterials](#) (INVASSAT, 50 h)
[Chemical sector](#) (INVASSAT, 50 h)
[Emergencies](#) (INVASSAT, 70 h)
[Nutrition](#) (INVASSAT, 50 h)
[Educational](#) (INVASSAT, 50 h)
[Services](#) (INVASSAT, 50 h)
[Researcher](#) (UPV, 15 h)

Transversal competences

[Gender perspective](#) (EVES, 20 h)
[Teamwork](#) (Labora, 25 h)
[Design Thinking](#) (Labora, 25 h)
[Critical Thinking](#) (Labora, 25 h)
[Adaptability, flexibility and agility](#) (Labora, 25 h)
[Workplace autonomy and innovation, initiative and creativity](#) (Labora, 25 h)
[Improving professional efficiency](#) (Labora, 25 h)
[Entrepreneur gender perspective](#) (UPV, 20 h)

Labor/Industrial

[Op. Industrial trucks](#) (Gescoform, 15 h)
[Food Hygiene Course](#) (Asonoman, 30 h)
[Self-protection plans](#) (INVASSAT, 15 h)
[Static electricity](#) (INVASSAT, 15 h)
[Data privacy](#) (CSIRT-CV, 10 h)

Information technology

[Using Python for Research](#) (HarvardX, 50 h)
[Analyzing Data With Python](#) (IBM, 20 h)
[Visualizing Data with Python](#) (IBM, 20 h)

PUBLICATIONS

Journal articles

[Loudspeakers for Vented Enclosures: a Backwards Approach for Speaker Selection \(LoVE BASS\)](#) · [Voice Coil](#), 2019

Conference papers

[Perfect broadband sound absorber metamaterial for noise reduction in a rocket launch](#) · [European Congress and Exposition on Noise Control Engineering \(EuroNoise 2021\)](#), 2021
[Sound diffusing metasurfaces based on elastic plates and membranes](#) · [European Congress and Exposition on Noise Control Engineering \(EuroNoise 2021\)](#), 2021
[Application of metamaterials to control noise scattering during space vehicle lift-off](#) · [European Congress and Exposition on Noise Control Engineering \(EuroNoise 2021\)](#), 2021
[Launch sound level characterisation and mitigation: numerical modelling framework and metamaterial proof of concept](#) · [16th European Conference on Spacecraft structures, Materials & Environmental Testing](#), 2021
[Acoustic field prediction during the launch of rockets](#) · [Tecnicaústica 2020: 50º Congreso](#)

phonometry

106 ★ • 27 releases • PyPI

Python • DSP • Acoustics

Standards-conformant acoustics library: sound level metrology, psychoacoustics, room, building, environmental and underwater acoustics, vibration, electroacoustics and wave simulation — every method implemented from the standard, with 550 CI-enforced conformance checks across 367 standards. Formerly PyOctaveBand.

cs-routeros-bouncer

14 ★ • 14 releases • 413 downloads • On the CrowdSec Hub

Go • Firewall • MikroTik • WAF

CrowdSec bouncer for MikroTik RouterOS: feeds the collaborative IPS/WAF CrowdSec decisions into the RouterOS firewall (address lists) through its API, blocking malicious IPs in real time.

Cloudflare-DNS-Updater

28 ★ • 7 releases • 104 downloads

Bash • DevOps

Dynamic DNS client for Cloudflare (IPv4/IPv6) written in Bash, with a bats test suite and coverage reporting.

TFG-TFM_EPS

84 ★ • 5 releases • 23 downloads

LaTeX • Academic template

LaTeX template for Bachelor's and Master's theses at the University of Alicante (EPS), with an automated CI build pipeline.

// EDUCATION

PhD program in Health and Wellness Technologies (discontinued)

2019–2023

Polytechnic University of Valencia • Valencia, Spain

Master's Degree in Acoustical Engineering

Polytechnic University of Valencia • Gandia, Spain

Degree in Telecommunication Engineering in Sound and Image

University of Alicante • Alicante, Spain

Higher Vocational Degree — Superior Sound Technician

IES Luis García Berlanga • Alicante, Spain

Intermediate Vocational Degree — Electronics Technician

IES Salesianos (San José Artesano) • Elche, Spain

Español de Acústica. XI Congreso Ibérico de Acústica. Faro, Portugal, 2020

Beyond Schroeder diffusers using acoustic metasurfaces • Tecniacústica 2020: 50º Congreso Español de Acústica. XI Congreso Ibérico de Acústica. Faro, Portugal, 2020

Aprendizaje basado en proyectos en las materias transductores acústicos y vibroacústica •

Memorias del Programa de Redes-ICE de calidad, innovación e investigación en docencia universitaria. Convocatoria 2017-18, 2018

Cálculo corregido, basado en la teoría moderna, de los campos acústicos (directo, temprano y tardío) • Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica, 2018

Campo directo (útil)/reverberado (perjudicial) resultados experimentales frente a simulación en EASE • Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica, 2018

Comportamiento vibroacústico de contenedores cilíndricos en aire • Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica, 2018

Campo directo (útil)/reverberado (perjudicial) resultados experimentales frente a simulación en catt-acoustic • Tecniacústica 2017: 48º Congreso Español de Acústica; Encuentro Ibérico de Acústica; European Symposium on Underwater Acoustics Applications; European Symposium on Sustainable Building Acoustics, 2017

Theses

Difusores acústicos basados en resonadores de membrana y placa • Universitat Politècnica de València. Departamento de Física Aplicada. Escuela Politécnica Superior de Gandia, 2019

Estudio de la relación campo directo/reverberado; útil/perjudicial • Universidad de Alicante. Departamento de Física, Ingeniería de Sistemas y Teoría de la Señal, 2018